

DSA STUDY BLUEPRINT SERIES

90. Flatten Binary Tree to Linked List

Converting Trees to Linked List structures

Section 09 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Flatten:** Rewire node pointers such that tree becomes a right-skewed linked list.
- Follow preorder sequence traversal in-place.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
void flatten(TreeNode* root) {  
    TreeNode* curr = root;  
    while (curr) {  
        if (curr->left) {  
            TreeNode* pred = curr->left;  
            while (pred->right) pred = pred->right;  
            pred->right = curr->right; curr->right = curr->left; curr->left = nullptr;  
        }  
        curr = curr->right;  
    }  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	O(N)
Space Complexity	O(1) (Iterative Morris)

Key Practices & Gotchas

- **Important:** Clean up left pointer properties (`left = nullptr`) after rewriting linkages.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace recursive Preorder Traversal visits (Node-Left-Right) for tree `1 (left=2, right=3)`:

Call Stack Level	Active Node Visited	Expected Left Traversal	Expected Right Traversal
1	Root (1)	Recurse on Left child (2)	Recurse on Right child (3)
2	Leaf (2)	Base case (nullptr)	Base case (nullptr)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Iterative traversals:** Standard DFS traversals can run iteratively by simulating recursion calls with Stacks.
- **Level Order BFS:** Level tracking searches use Queues to visit node layers.
- **Related LeetCode Problems:** #102 Binary Tree Level Order Traversal, #94 Binary Tree Inorder Traversal, #236 Lowest Common Ancestor.